

1. General Use Case Diagram

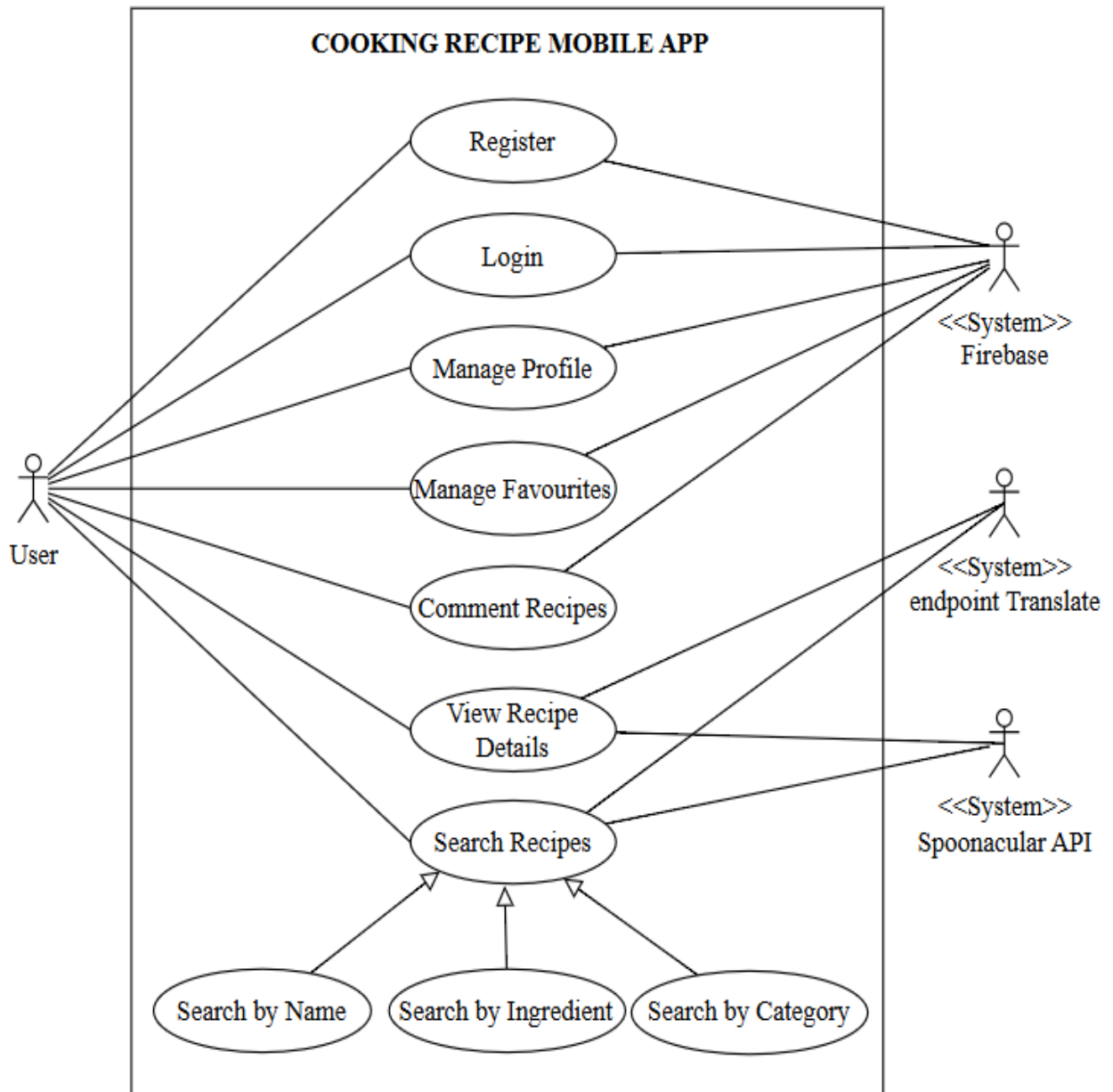


Figure 1. General Use Case diagram of the recipe search application system.

2. Account Registration Function

2.1. Use Case Diagram

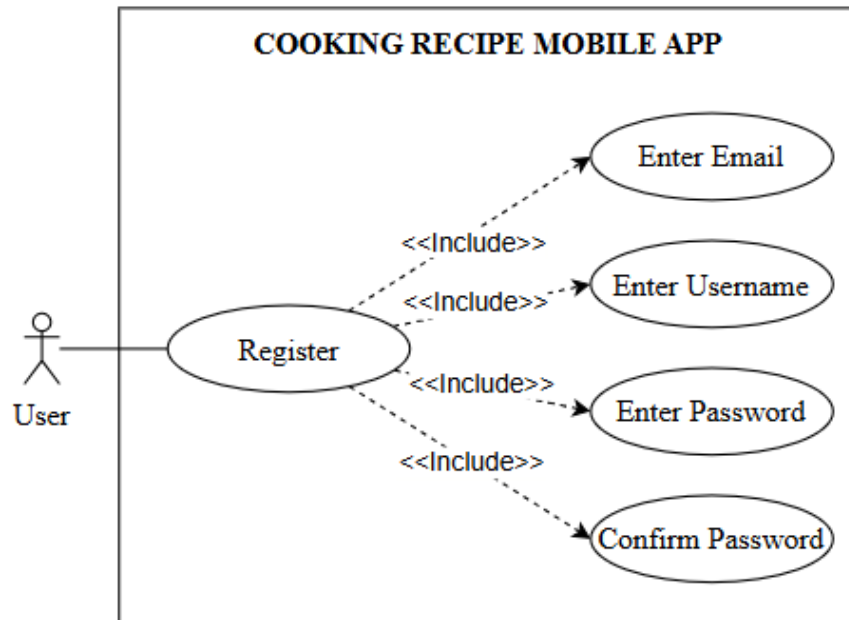


Figure 2. Use Case diagram for the account registration function.

2.2. Use Case scenario

Use case ID	UC001		
Use case name	Account registration.		
Actor	User.		
Description	Allows the user to create a new account using Email, Username, and Password to use the application.		
Preconditions	The user is currently on the Registration screen.		
Main flow of events	Step	Performed by	Action
	1	User	Enters Email, Username, Password, re-enters password, and presses Register.
	2	Firestore Authentication	Check the validity of the data

	3	Firestore Authentication	Checks whether the Email and Username already exist.
	4	System	If valid: Creates an account using Firestore Authentication
	5	System	Saves user information into Cloud Firestore.
	6	System	Displays a successful registration notification and navigates to the home page.
Alternative flows	Step	Performed by	Action
	3a	System	Data is invalid, reports error and requests re-entry.
	4a	System	Username exists, notifies that the Username already exists.
	4b	System	Account creation failed, displays an error notification.
	5a	System	Data saving failed. Cancels the newly created account and notifies the error.
Postconditions	The account is successfully created, information is saved on the system, and the user is redirected into the application.		

Table 1. Use Case scenario for the account registration function.

3. System Login Function

3.1. Use Case Diagram

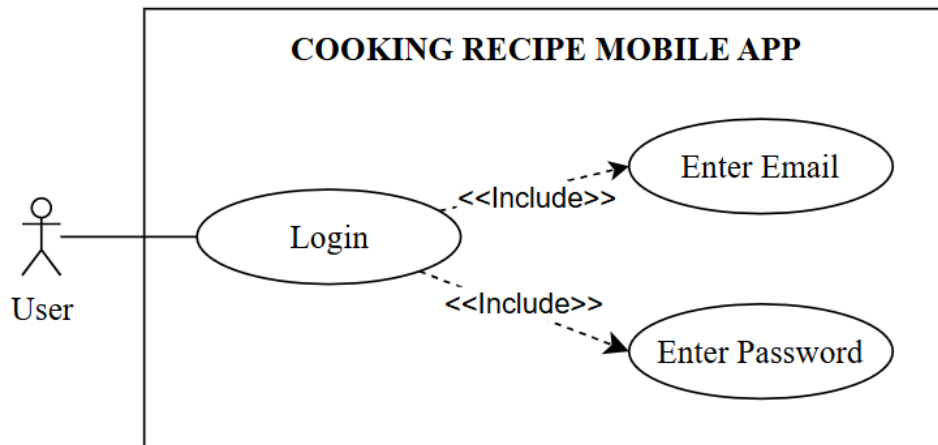


Figure 3. Use Case diagram for the system login.

3.2. Use Case scenario

Use case ID	UC002		
Use case name	System login.		
Actor	User.		
Description	Allows the user to log in using Email and Password to access the application's functions.		
Preconditions	The user already has an account and is currently on the Login screen.		
Main flow of events	Step	Performed by	Action
	1	User	Enters Email, Password, and presses Log In.
	2	Firestore Authentication	Checks the validity of the entered data.
	3	System	Authenticates information using Firestore Authentication.
	4	System	Successful authentication, loads user information from Firestore.

	5	System	Notifies successful login and navigates to the Home page.
Alternative flows	Step	Performed by	Action
	2a	System	Entered data is invalid. Displays an error notification and requests re-entry.
	3a	System	Incorrect Email or Password. Notifies login failure.
	3b	System	Connection error or system error. Displays a notification and allows logging in again.
Postconditions	The user logs in successfully, the login session is saved, and the user is redirected to the Home page.		

Table 2. Use Case scenario for the system login.

4. Personal Information Management Function

4.1. Use Case Diagram

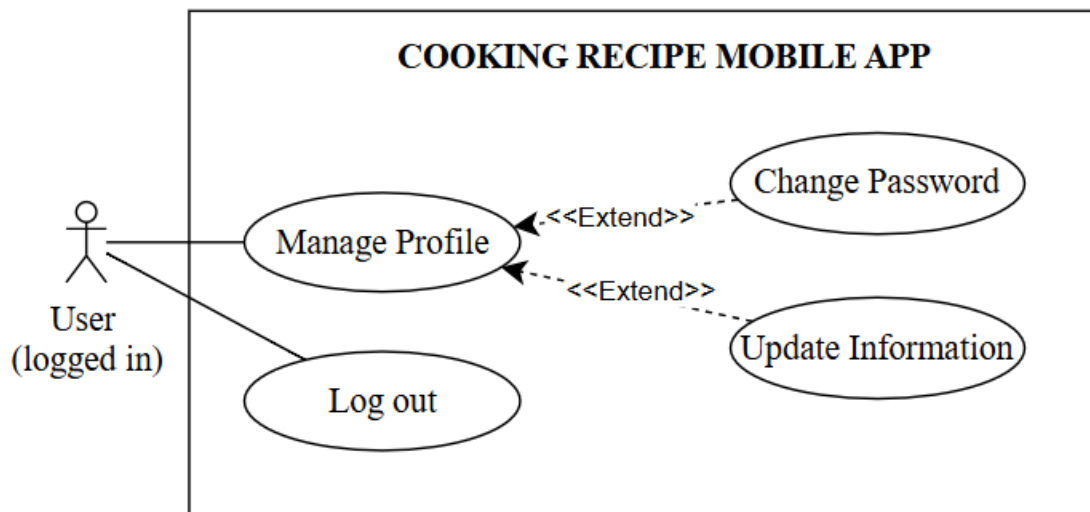


Figure 4. Use Case diagram for the account information management function.

4.2. Use Case scenario

Use case ID	UC003		
Use case name	Account management.		
Actor	User (logged in).		
Description	Allows the user to view, update account information, change password, or log out of the system.		
Preconditions	The user has logged in and is currently on the Account screen.		
Main flow of events	Step	Performed by	Action
	1	User	Accesses the Account Management section.
	2	System	Displays the current account information.
	3	User	Selects one of two functions: - Updates email, username, or changes password and presses Save changes - Presses Log Out
	4	System	Processes the request and updates the corresponding data.
	5	System	Displays the result notification to the user.
Alternative flows	Step	Performed by	Action
	3a	System	Updated information is invalid. Displays an error notification.
	4a	System	Updating email, username, or changing password failed.

			Notifies the error and requests re-execution.
	3b	System	Successful logout. The system clears the login session and redirects to the Login screen.
Postconditions	Account information is successfully updated or the user has logged out of the system.		

Table 3. Use Case scenario for the account information management function.

5. Recipe Search Function

5.1. Use Case Diagram

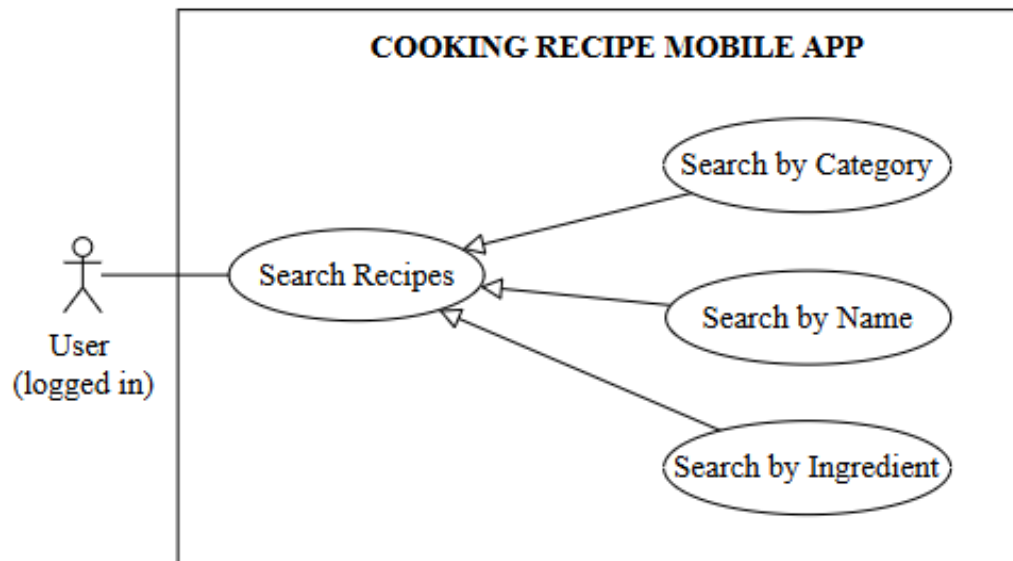


Figure 5. Use Case diagram for the recipe search function.

5.2. Use Case scenario

Use case ID	UC004
Use case name	Recipe Search
Actor	User (logged in).
Description	Allows the user to search for dishes by keywords or filter by category to view suitable recipes.

Preconditions	The user is currently on the Home screen or the Dish Category screen.		
Main flow of events	Step	Performed by	Action
	1	User	Can - Search by Category - Search by dish's name - Search by Ingredient
	2	System	Receives the request and sends a query to the Spoonacular API.
	3	Spoonacular API	Returns a list of matching dishes.
	4	System	Processes data and updates the dish list on the interface.
	5	System	Displays search or filter results to the user.
Alternative flows	Step	Performed by	Action
	1a	System	User does not enter a keyword, displays the default list.
	3a	System	No matching dishes found. Displays a no results notification.
	3b	System	Network connection error or API error. Displays an error notification and allows the user to re-execute the operation.
Postconditions	The dish list is displayed according to the search keyword or category that the user selected.		

Table 4. Use Case scenario for the recipe search function.

6. Recipe Detail Function

6.1. Use Case Diagram

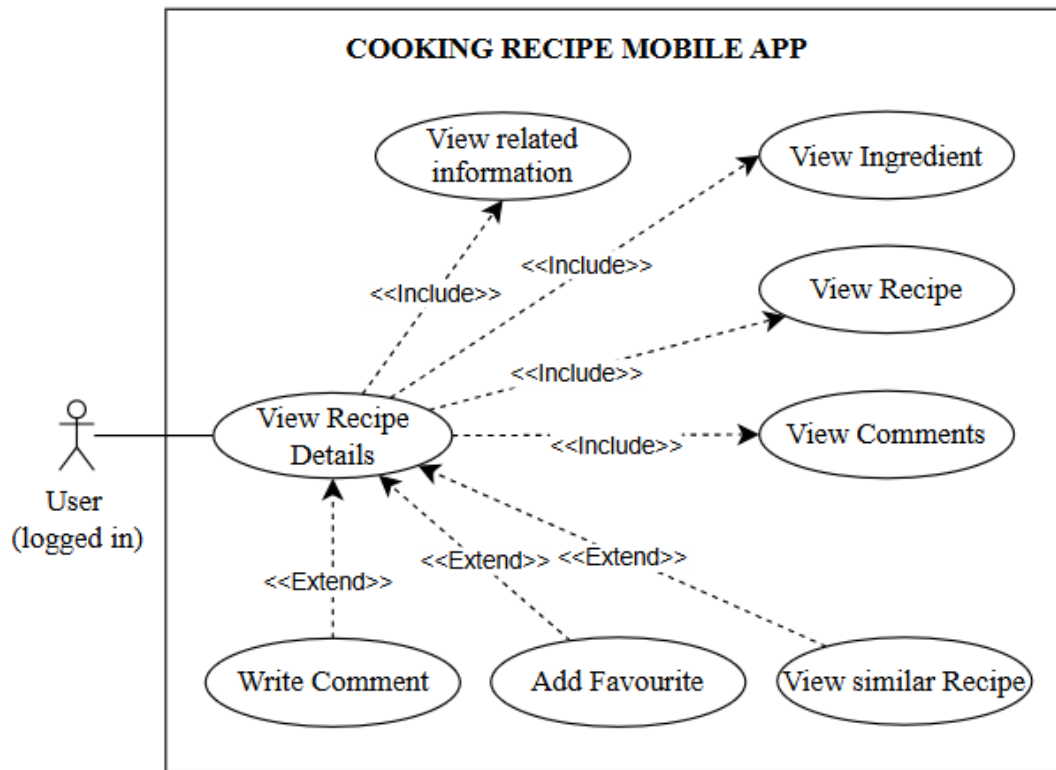


Figure 6. Use Case diagram for the recipe detail function.

6.2. Use Case scenario

Use case ID	UC005		
Use case name	View recipe details.		
Actor	User (logged in).		
Description	Allows the user to view detailed information of a dish, including ingredients, cooking instructions, cooking time, calorie count, and related information.		
Preconditions	The user has logged in and is currently viewing a list of dishes.		
Main flow of events	Step	Performed by	Action
	1	User	Selects a dish from the list.

	2	System	Sends a query to the Spoonacular API.
	3	Spoonacular API	Returns the corresponding recipe and ingredients.
	4	System	Displays detailed information of the recipe, ingredients, and cooking instructions.
	5	User	Can: - View related information. - View comments and write comments. - Add to favorites. - View similar dishes.
Alternative flows	Step	Performed by	Action
	2a	System	Dish data not found. Displays an error notification.
	4a	System	Network connection or API error. Displays a notification and allows reloading data.
Postconditions	Detailed information of the dish is fully displayed on the interface for the user to reference and interact with.		

Table 5. Use Case scenario for the recipe detail function.

7. Favourite Recipe Management Function

7.1. Use Case Diagram

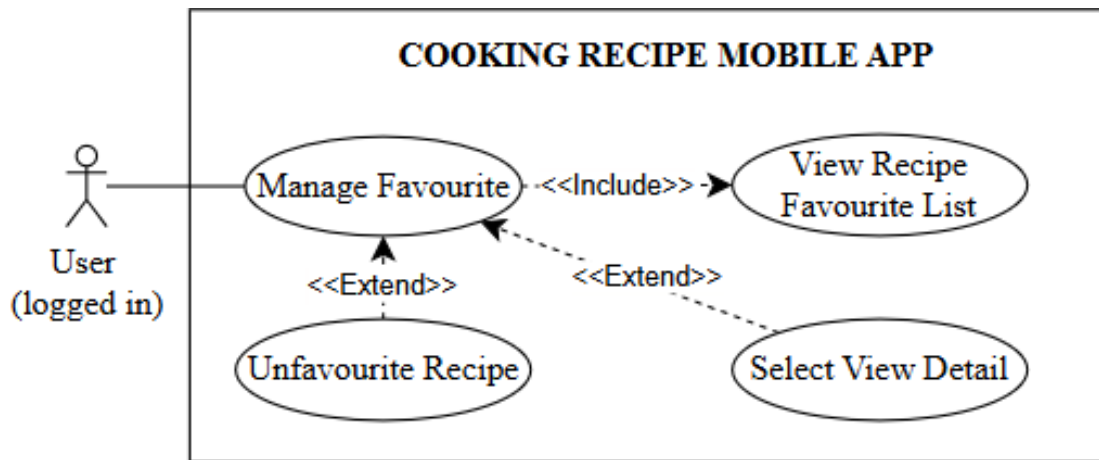


Figure 7. Use Case diagram for the favourite recipe management function.

7.2. Use Case scenario

Use case ID	UC006		
Use case name	Manage favorite recipes.		
Actor	User (logged in).		
Description	Allows the user to view and manage the list of recipes that have been marked as favorites.		
Preconditions	The user has logged into the system.		
Main flow of events	Step	Performed by	Action
	1	User	Accesses the Favorite Recipes function.
	2	System	Queries the favorite recipe list from the Firebase Firestore database.
	3	Firebase Firestore	Queries the list and returns data to the system.
	4	System	Displays the list of dishes that have been marked as favorites.

	5	User	Can: View and select recipes, or Unfavorite.
Alternative flows	Step	Performed by	Action
	4a	System	No favorite recipes exist. The system displays a "No favorite dishes yet" notification.
	4b	System	Connection error or data retrieval failed. The system displays an error notification and allows reloading data.
Postconditions	The favorite recipe list is successfully displayed and updated according to the user's actions.		

Table 6. Use Case scenario for the the favourite recipe management function.

8. Comment Recipes Function

8.1. Use Case Diagram

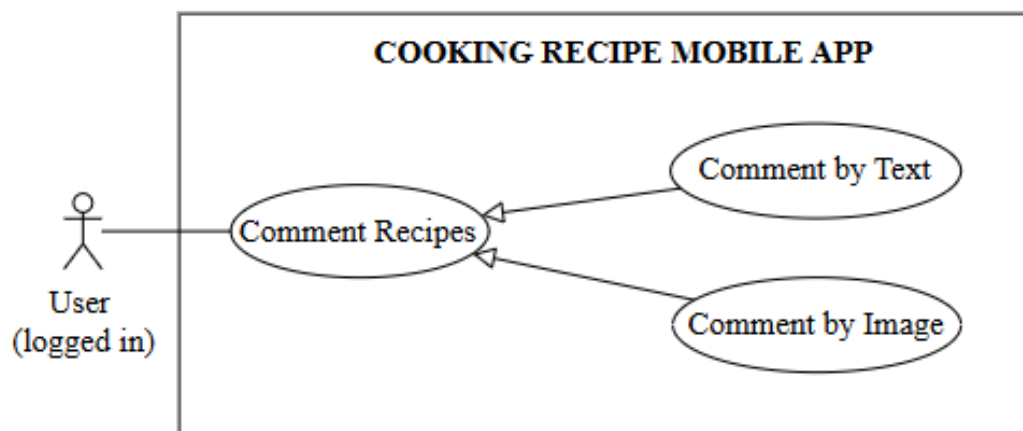


Figure 8. Use Case diagram for the comment recipes function.

8.2. Use Case scenario

Use case ID	UC007
Use case name	Comment Recipes
Actor	User (logged in).

Description	Allows the user to provide feedback and review a recipe by posting text comments or images		
Preconditions	The user has logged into the system and is currently viewing a recipe detail screen.		
Main flow of events	Step	Performed by	Action
	1	User	Enter text and/or attach an image, then tap “send” button.
	2	System	Validate the input data (ensure it is not empty and conforms to format rules).
	3	System	Save the comment data (text, image URL, user information, and timestamp) into the database.
	4	System	Refresh the comment section and display the newly added comment at the top of the list.
Alternative flows	Step	Performed by	Action
	2a	System	If the text and image are both empty, display "Comment cannot be empty" and stop processing.
	3a	System	If saving fails, display "Failed to post comment. Please try again" and preserve user input.
Postconditions	The user's comment is successfully posted and visible to all users viewing this recipe.		

Table 7. Use Case scenario for the the comment repices function.